



Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion

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Background - Perception in Winter

- Sensor Degradation

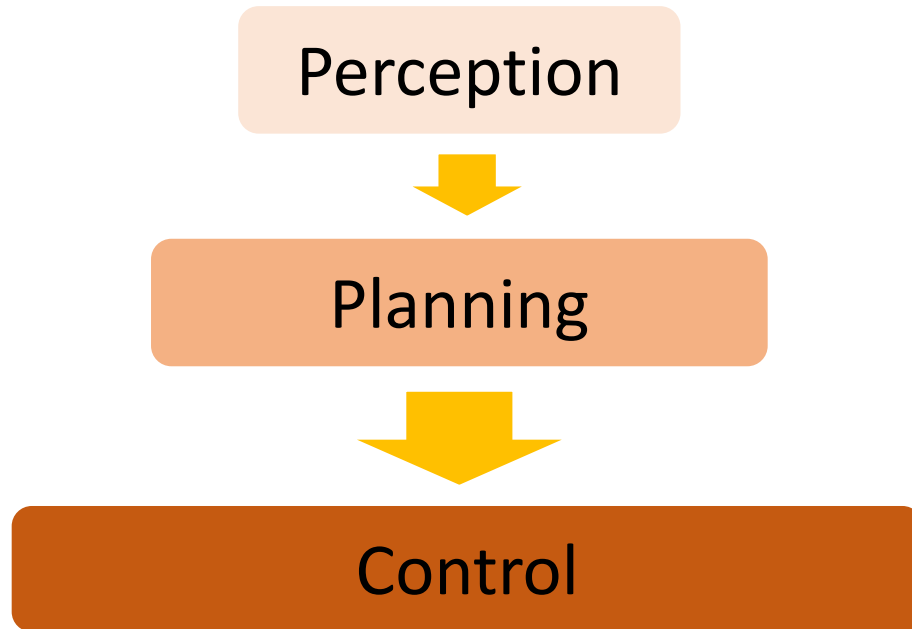


- Feature Loss



Background – Planning and Control in Winter

- Downstream Uncertainty



- Complex Dynamics



Winter Navigation Challenges

Snow impairs traditional navigation:

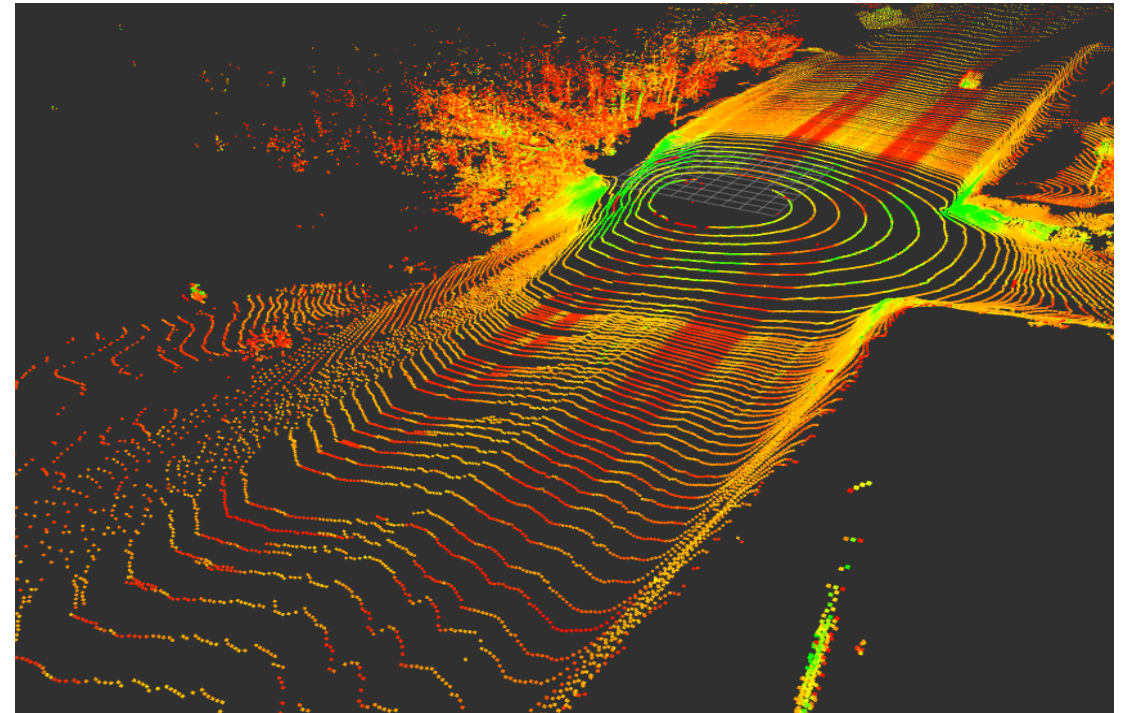
- High-definition (HD) maps [Sayed et al., 2023]
- LiDAR-SLAM [Pomerleau, 2022]
- Vision navigation [Baril et al., 2021]



Human Driving Strategy

How do human drivers drive – following wheel tracks:

- Verified [Kalra et al., 2023]
- More friction [Rinne et al., 2024]
- Extra lateral force [Ordonez et al., 2020]



Existing Research – Vision-based



CNN [Goberville et al, 2022]



YOLOv5 [Kim et al, 2024]



Rule-based [Zhang et al, 2024]

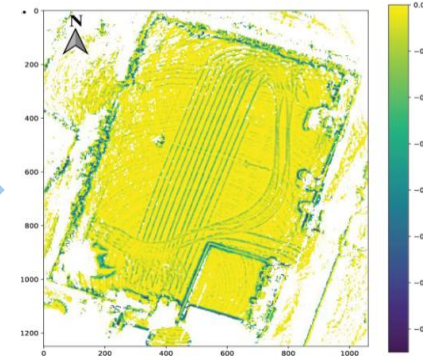
Existing Research - Geometry-based



Platform

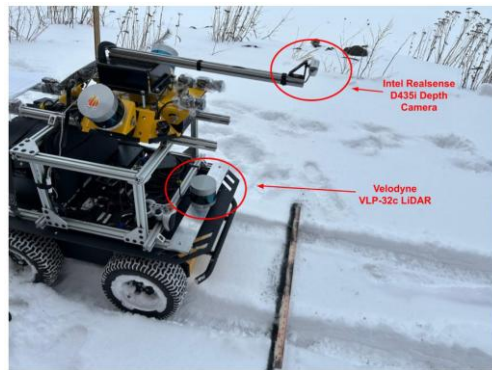


Experiment Setup

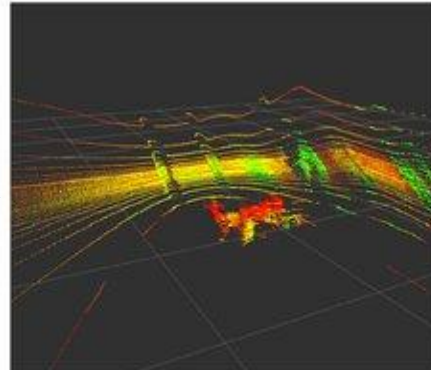


Rut Depth Map

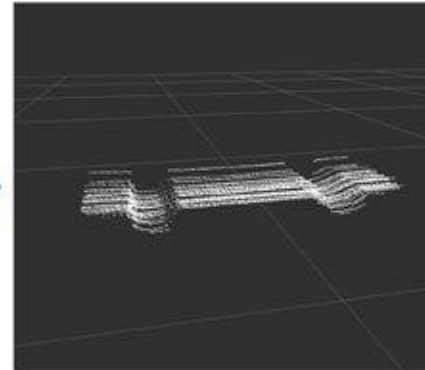
LiDAR-derived digital elevation models [Fuentes et al., 2021]



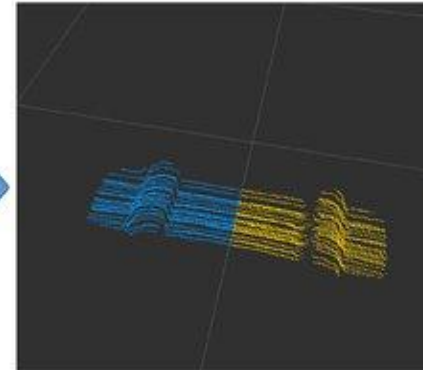
Platform



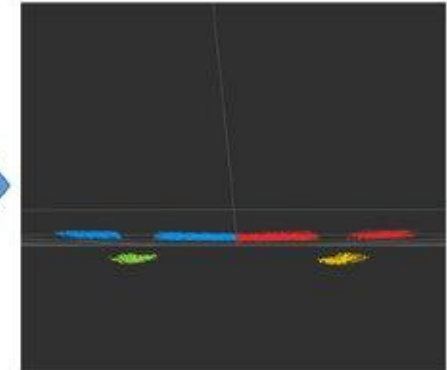
Input Data



Crop Background



Sort as left & right



Planar Segmentation

LiDAR + Depth camera for rut depth extraction [Ian et al., 2023]

Research Gap – Traversability Estimation in Featureless Environment



- Whiteout
- Occluded
- Low contrast
- Track degradation

Thermal Imaging in Winter

- RGB image



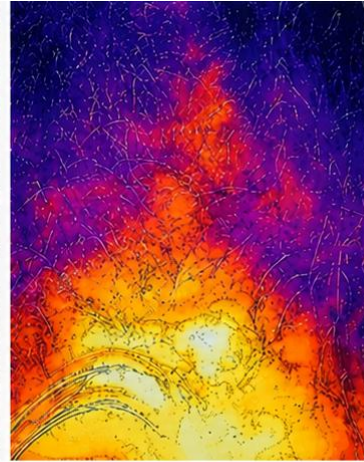
- Thermal image



Thermal Imaging



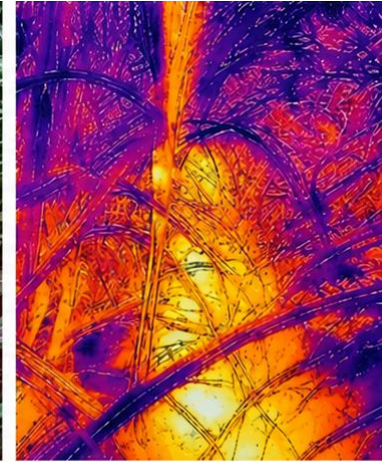
RGB



IR



RGB



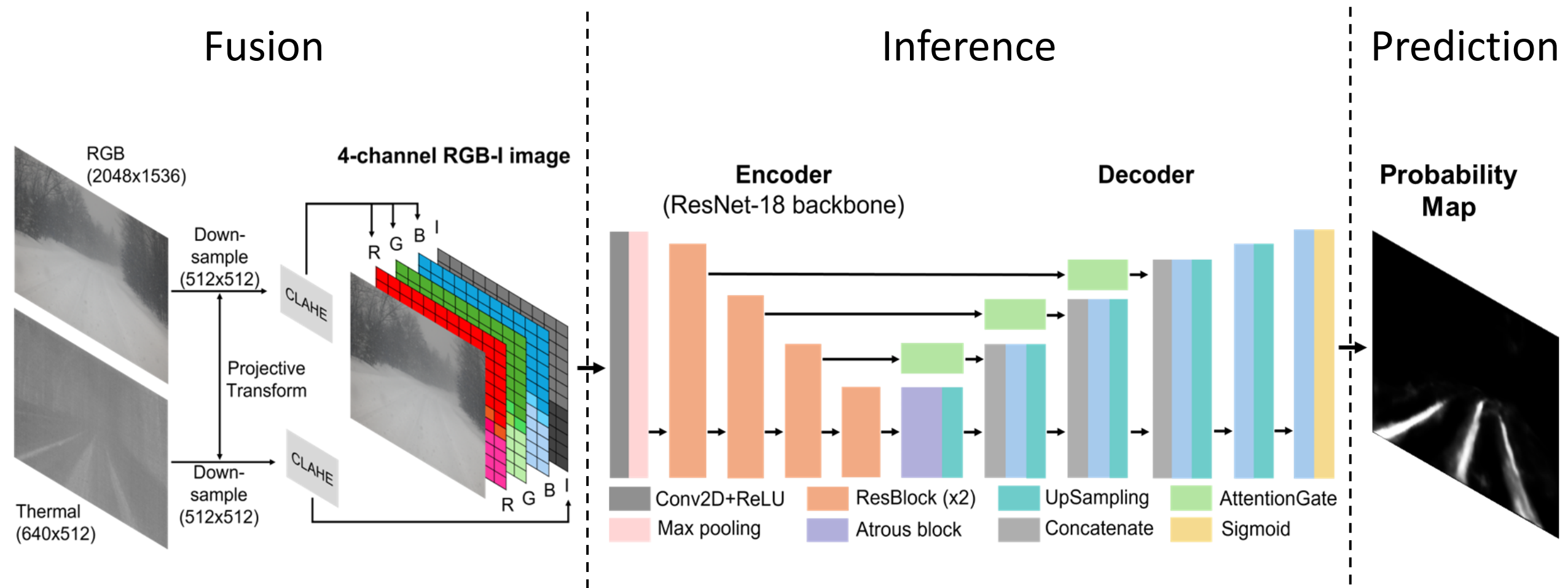
IR

Drivable area detection in sugarcane plantations [Xaud, et al., 2019]

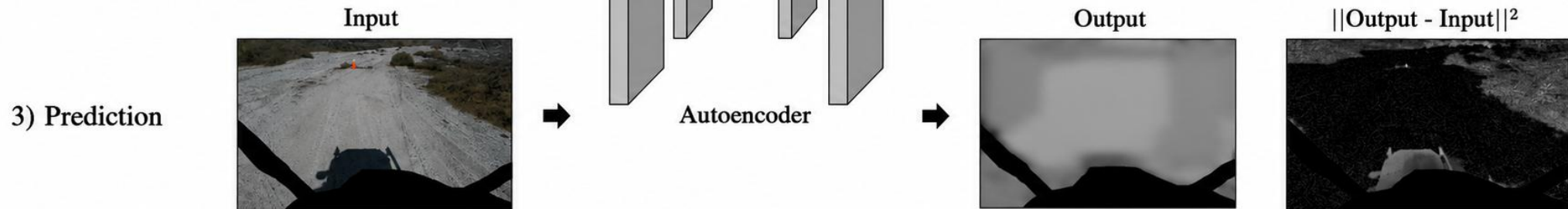
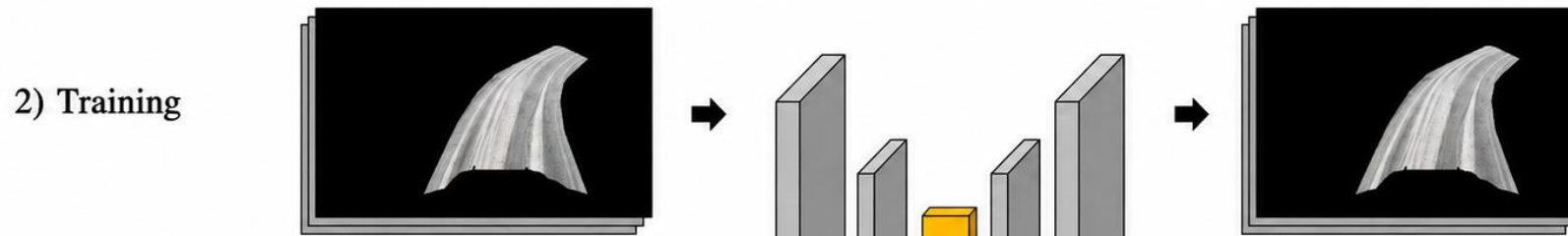
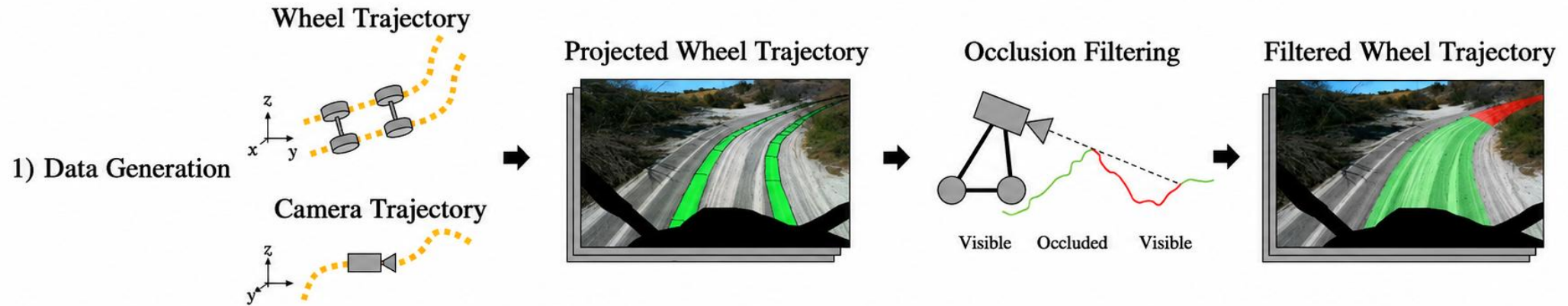


Path planning in urban area at night [Aditya, et al., 2024]

Proposed Approach

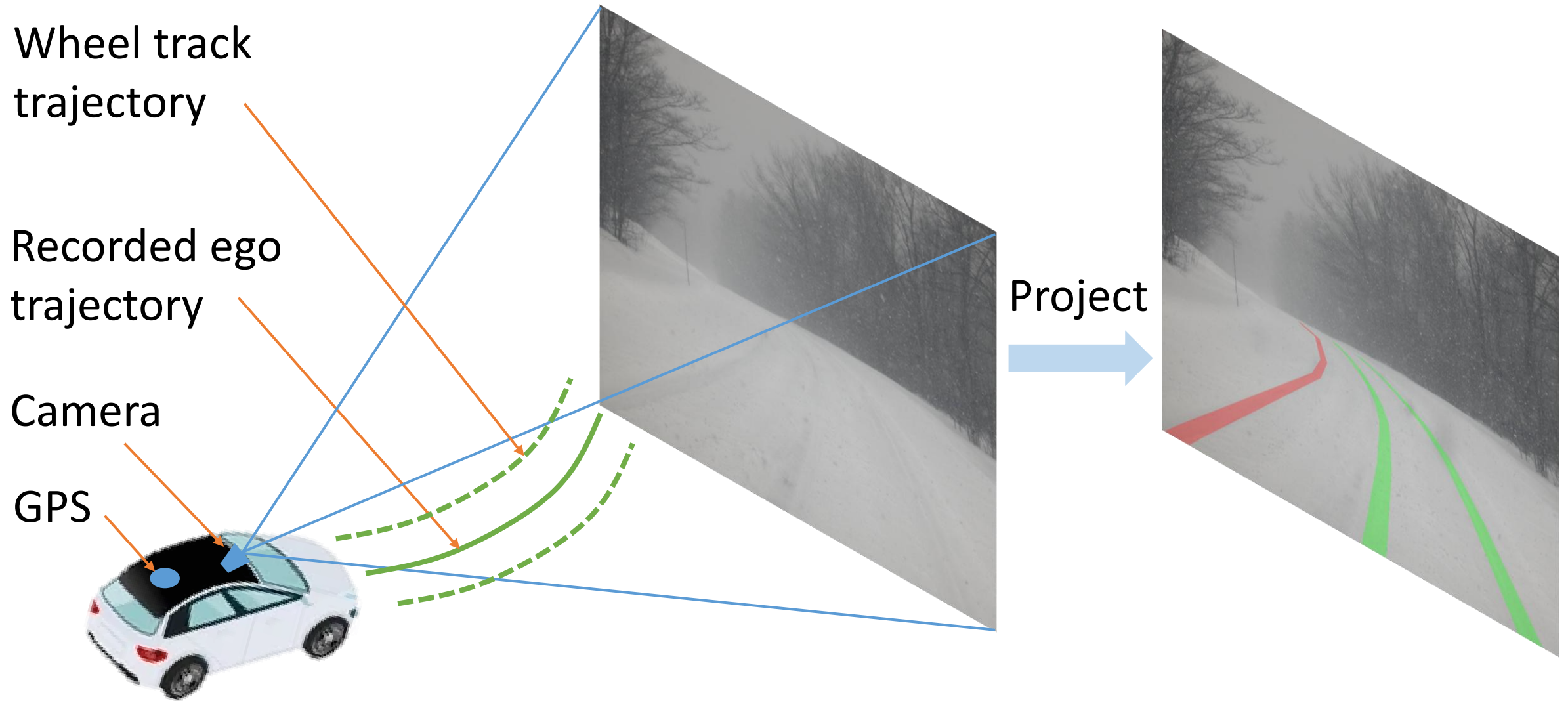


Self-supervised Traversability Estimation



Off-road traversability estimation [Schmid et al., 2022]

Self-supervised Labeling



Baseline Models

Name	Description
U-Net	Excellent for capture fine-grained details
PSPNet	Captures the big picture (global context)
DeepLabv3+	The current industry standard for autonomous driving
ResNet18	Used as the foundation to measure my design

Results

Model	AUROC	MaxF	AP	PRE	REC	FPR	FNR
UNet	0.7068	0.5197	0.6079	0.7691	0.4192	0.0056	0.5808
PSPNet	0.7532	0.6102	0.6666	0.7978	0.5122	0.0057	0.4878
DeepLabv3+	0.7446	0.6002	0.6629	0.8074	0.4943	0.0052	0.5057
ResNet18	0.8462	0.7202	0.7337	0.7499	0.7034	0.0109	0.2966

Overall Model Capability

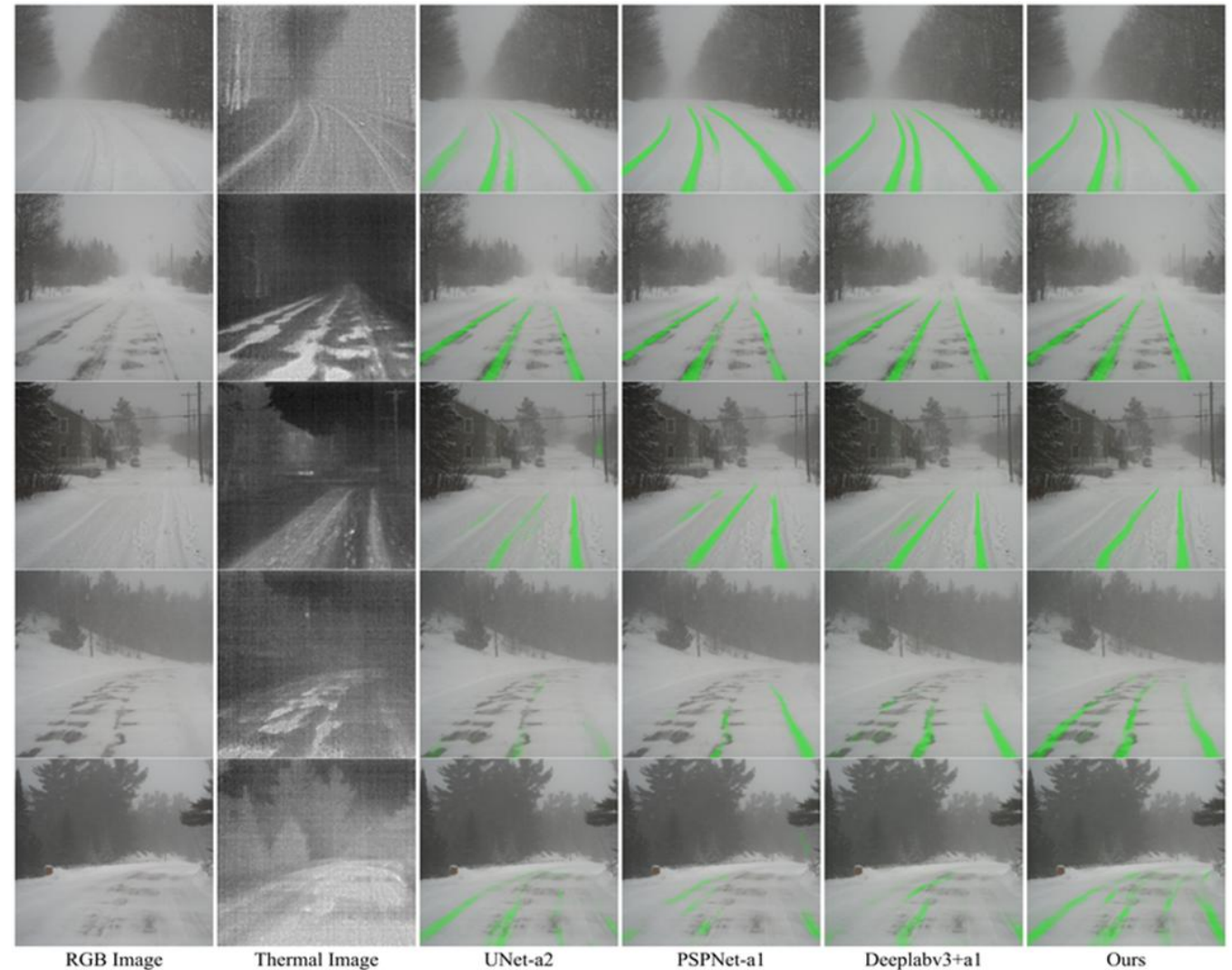
- **AUROC**: Area Under the Receiver Operating Characteristic Curve
- **AP**: Average Precision

Detection Completeness & Missed Signals

- **REC**: Recall / Sensitivity / True Positive Rate)
- **FNR**: False Negative Rate

Detection Precision & False Alarms

- **MaxF**: Maximum F1-score
- **PRE**: Precision
- **FPR**: False Positive Rate



Ablation Study

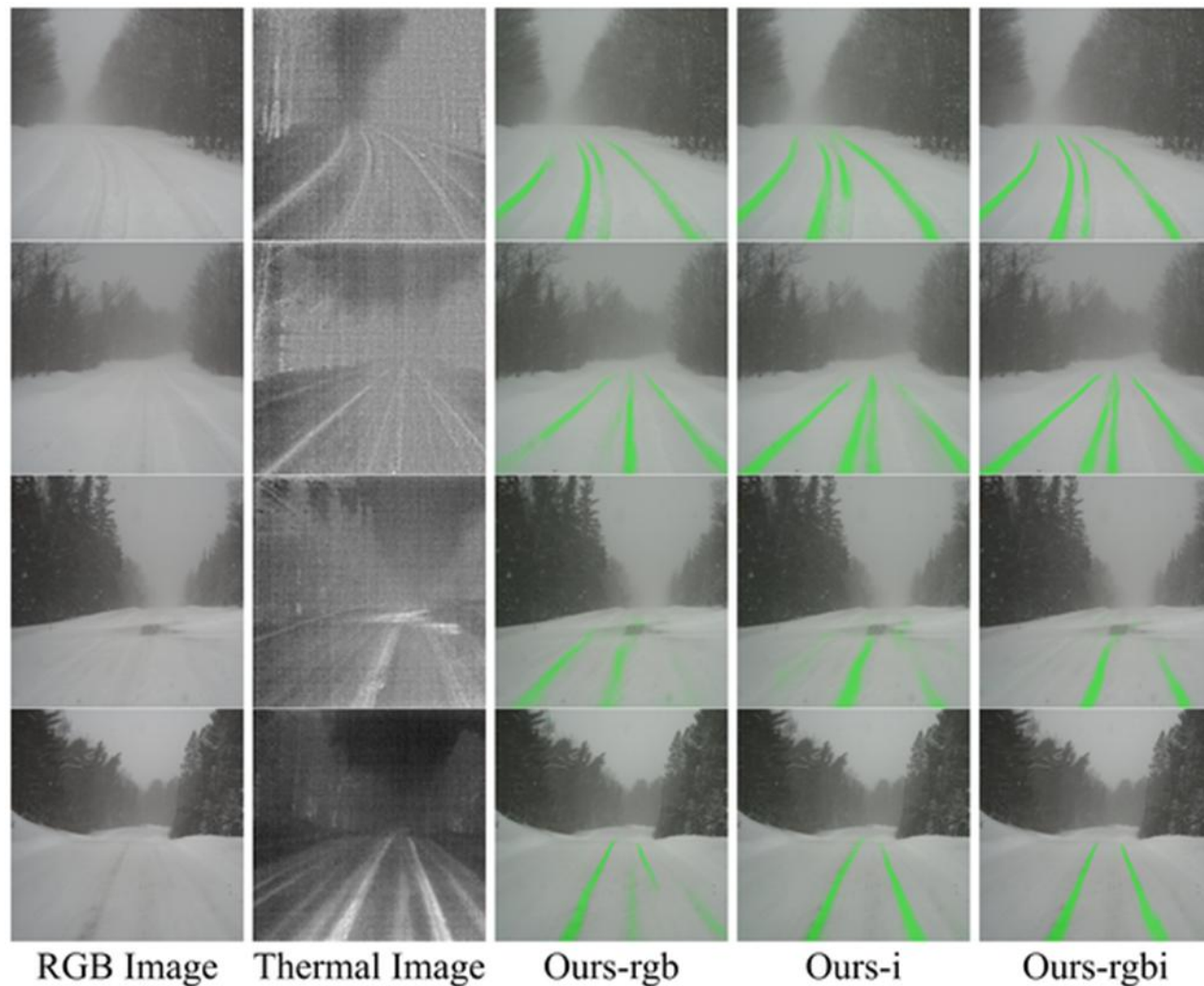
Model	AUROC	MaxF	AP	PRE	REC	FPR	FNR
UNet+a2	0.7843	0.6529	0.6941	0.7929	0.5753	0.0068	0.4247
PSPNet+a1	0.8476	0.7186	0.7337	0.7471	0.7064	0.0112	0.2936
DeepLabv3+a1	0.8589	0.7460	0.7584	0.7757	0.7281	0.0102	0.2719
Ours-rgb	0.6025	0.3083	0.5164	0.6901	0.2073	0.0024	0.7927
Ours-i	0.7687	0.5803	0.6127	0.6525	0.5510	0.0137	0.4490
Ours-rgbi	0.8744	0.7579	0.7675	0.7633	0.7601	0.0114	0.2399

Limitations of single-sensor modality:

- RGB-only (Ours-rgb) - suffers from an extreme Miss Rate (FNR: 79.27%)
- Thermal-only (Ours-i) - lacks sufficient spatial context

Cross-Modal (Ours-rgbi)

- Highest overall performance
- Low false negative rate

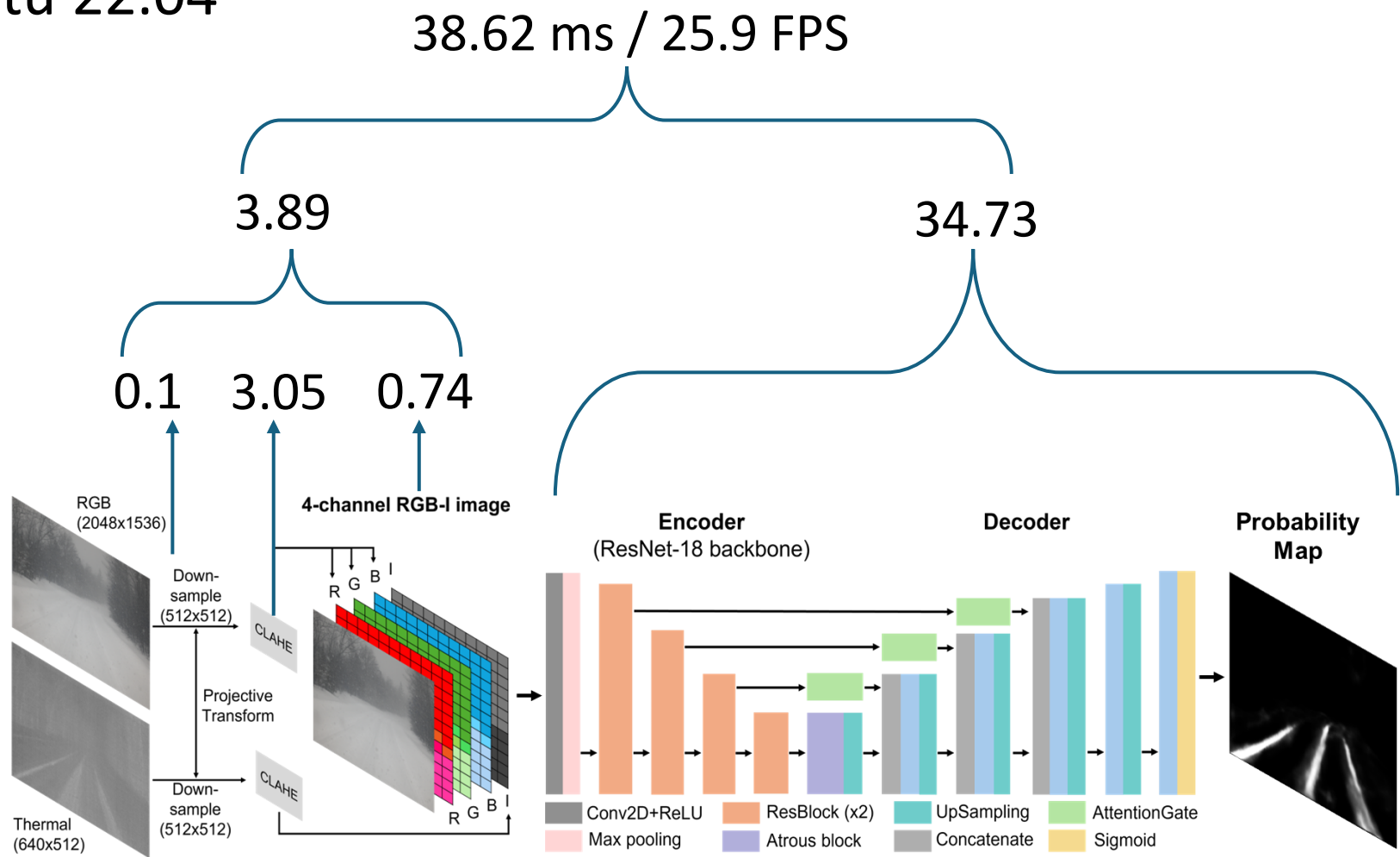


Latency & Throughput

NVIDIA RTX 5090 with Ubuntu 22.04

- Different stages
- Different models

Pipeline Stage / Model Variant	Latency (ms)	Throughput
<i>Preprocessing Pipeline</i>		
Stage 1: Downsample (RGB)	0.09	-
Stage 1: Downsample (Thermal)	0.01	-
Stage 2: CLAHE (RGB)	2.10	-
Stage 2: CLAHE (Thermal)	0.95	-
Stage 3: Merge	0.74	-
Total Preprocessing	3.89	257.1 Hz
<i>Model Inference (Stage 4)</i>		
UNet	23.43	42.7 FPS
UNet + a2	29.78	33.6 FPS
DeepLabv3+	26.33	38.0 FPS
DeepLabv3+a1	33.15	30.2 FPS
PSPNet	25.26	39.6 FPS
PSPNet+a1	30.54	32.8 FPS
ResNet18 (Baseline)	26.85	37.2 FPS
ResNet18+a2 (Proposed)	34.73	28.8 FPS
End-to-End System (Proposed)	38.62	25.9 FPS



Conclusion

- White-out & featureless winter traversability estimation.
- Thermal imaging for wheel track detection.
- A Thermal-RGB fusion pipeline based on ResNet-18.
- Thermal image improves wheel track detection:
 - Reducing the False Negative Rate by $\sim 70\%$ compared to RGB input.
 - Generating more complete and continuous wheel tracks.

Inspiration & Future Work

- Diverse terrains, e.g., Oklahoma:



Black Ice &
Flash-Freezes



Severe Weather –
Heavy Rain



Severe Weather –
Heavy Fog

- Generative AI / VLAs

Q & A